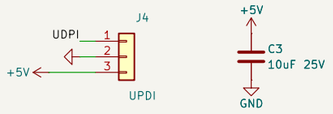
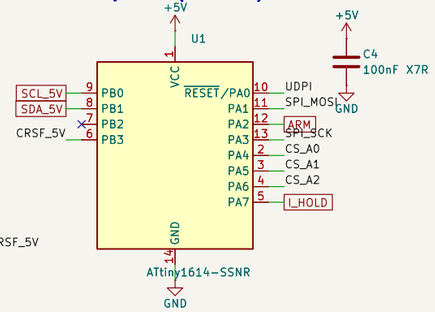
DAC SHIM: ATtiny1614 reads the MPU-6050 (off-board GY-521) and writes 5x MCP4822; every DAC output is RC filtered (1k/1uF, 160 Hz) before the PID sheets

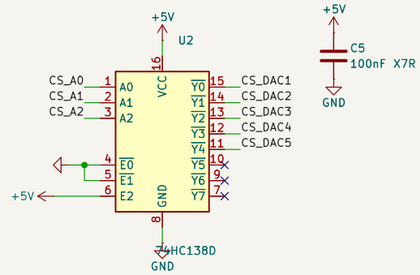
UPDI (Adafruit UPDI Friend)



MCU: ATtiny1614 (SOIC-14)

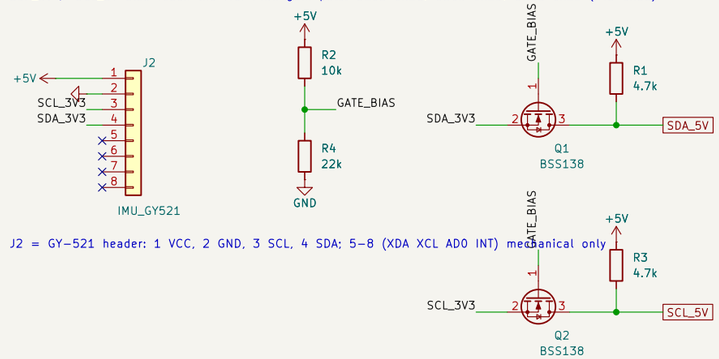


DAC chip-select decoder: 74HC138 (Y0..Y4 -> DAC1..5)



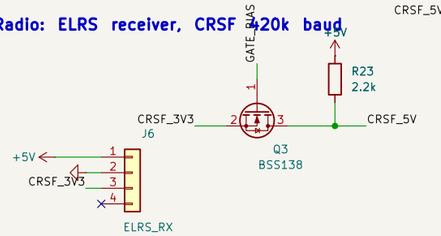
IMU: GY-521 site at the board centre (CG); BSS138 I2C level shifter

SDA_5V / SCL_5V also reach the MCP4451 gain pots: U203 0x2C, U303 0x2D, U103 0x2E (IMU 0x68)



J2 = GY-521 header: 1 VCC, 2 GND, 3 SCL, 4 SDA; 5-8 (XDA XCL ADO INT) mechanical only

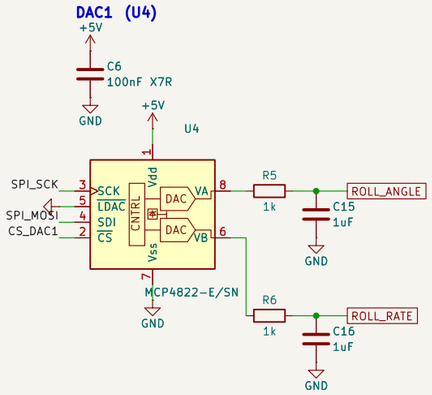
Radio: ELRS receiver, CRSF 420k baud



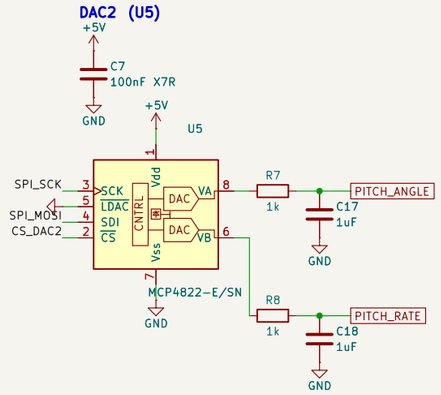
J6: 1 5V, 2 GND, 3 receiver TX, 4 receiver RX (unused)

DACs: MCP4822 (SOIC-8), SPI shared, CS from the decoder; outputs are 1k/1uF RC filtered

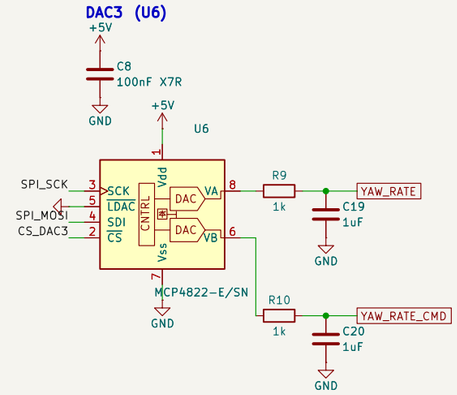
DAC1 (U4)



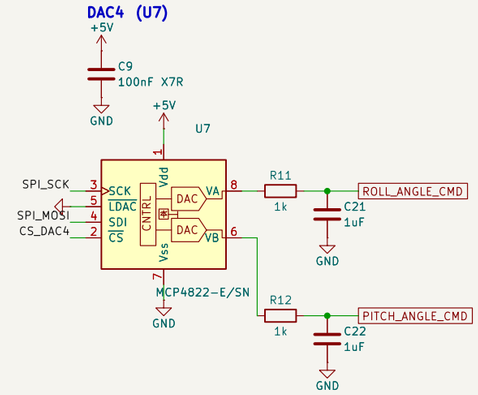
DAC2 (U5)



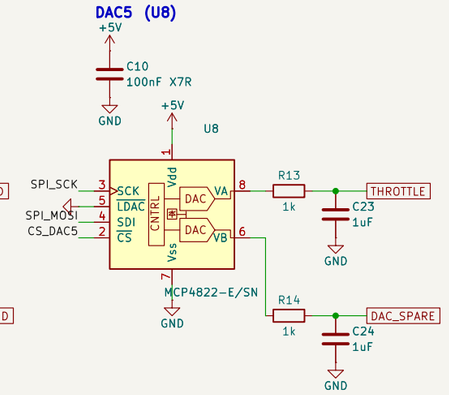
DAC3 (U6)



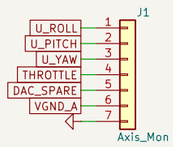
DAC4 (U7)



DAC5 (U8)



Axis monitor / scope header (relative to VGND_A)



Sub-sheets: analog PID + control I/O

PID_Reference
File: pid_reference.kicad_sch

PID_Roll
File: pid_roll.kicad_sch

PID_Pitch
File: pid_pitch.kicad_sch

PID_Yaw
File: pid_yaw.kicad_sch

Control_IO
File: control_io.kicad_sch

M3 holes: H1-H4 corners, H5-H8 20x20 ESC stack

H1 H2 H3 H4

H5 H6 H7 H8

Analog PID quad flight controller

Sheet: /
File: PID_Board_SMT.kicad_sch

Title: Quad flight controller (SMT): DAC shim + analog PID

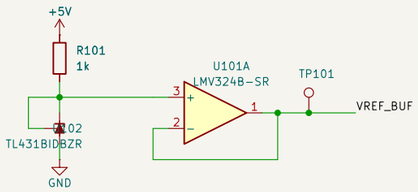
Size: A3	Date: 2026-10-20	Rev:
KiCad E.D.A. 10.0.3		Id: 1/6

PID REFERENCE: TL431 -> buffer -> VSET divider -> three isolated virtual grounds (VG = 2.048 V)

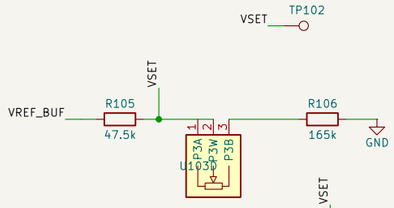
Set the U103 VSET channel (I2C) until (DAC code 2048) — VGND_A < 1 mV. Never tie VGND_A/B/C together. 100 ohm isolates each follower from its 470 nF rail cap.

TL431 reference

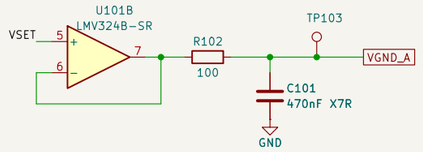
Buffer (U101A)

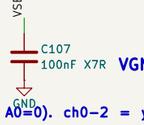


VSET divider (digital trim)

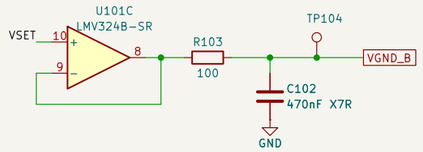


VGND_A follower (U101B)

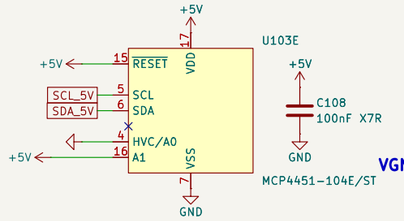




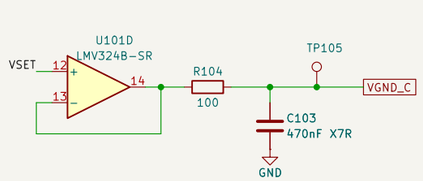
VGND_B follower (U101C)



U103 MCP4451: I2C 0x2E (A1=1, A0=0). ch0–2 = yaw P/I/D, ch3 = VSET



VGND_C follower (U101D)



U101 power, 5 V bulk



Analog PID quad flight controller

Sheet: /PID_Reference/
File: pid_reference.kicad_sch

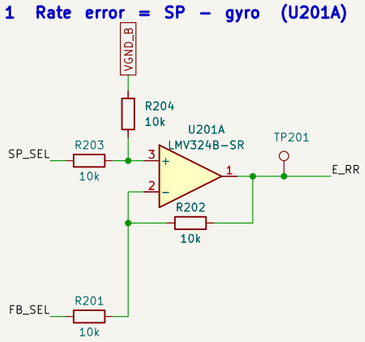
Title: PID reference + virtual grounds

Size: A3 | Date: 2026-10-20 | Rev:
KiCad E.D.A. 10.0.3 | Id: 2/6

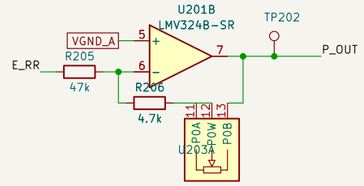
ROLL AXIS: rate PID + angle loop + bench plant (guide chips U2 = U201, U3 = U202)

Signals are deviations from VG = 2.048 V. Every P, I and D path is inverted twice. Gains are set by MCP4451 rheostat channels (A tied to W; R_BW = code/256 x 100k).

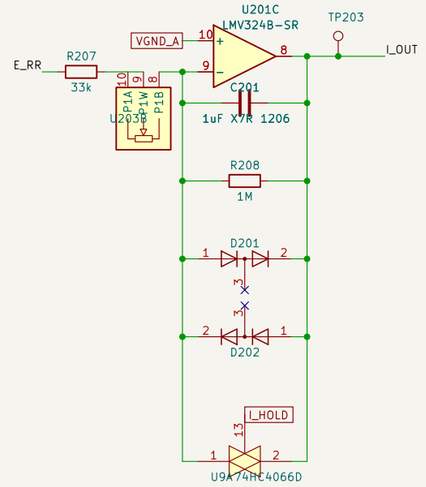
1 Rate error = SP − gyro (U201A)



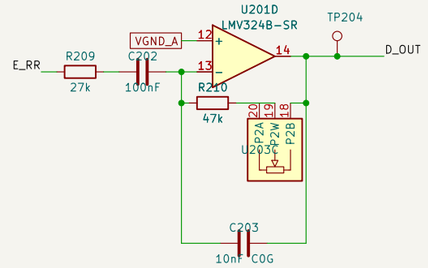
2 P, inverting (U201B)



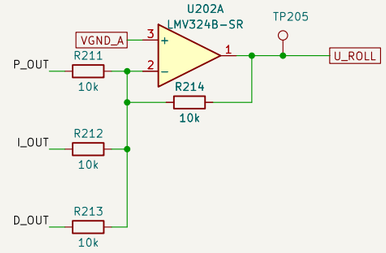
3 Leaky I + clamp + reset (U9 = 74HC4066) (U201C)



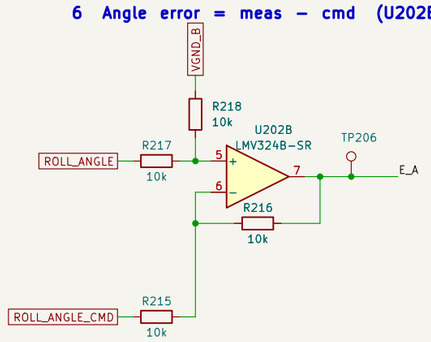
4 Filtered D (U201D)



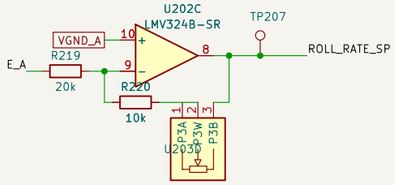
5 Summer -> U_ROLL (U202A)



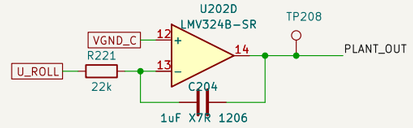
6 Angle error = meas − cmd (U202B)



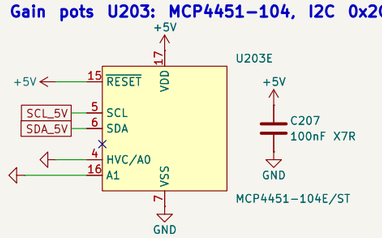
7 Angle P -> rate SP (U202C)



8 Bench plant (U202D)

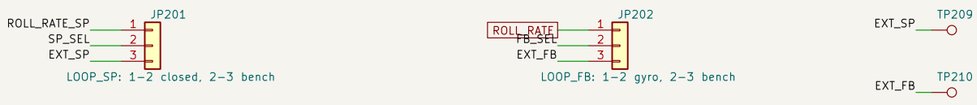


Gain pots U203: MCP4451-104, I2C 0x2C (A1=0, A0=0)



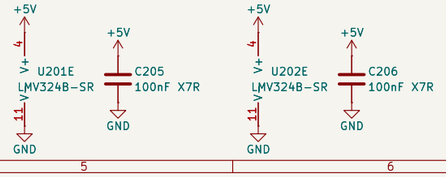
ch0 = P: code 128 = 50k (Kp 1.16)
ch1 = I: code 171 = 67k (Ki 10/s)
ch2 = D: code 128 = 50k (Kd 9.7 ms)
ch3 = Angle P: code 64 = 25k (Go 1.75)
Power-on = code 128 (mid); the shim ATtiny writes the stored codes over I2C at boot.

Loop select + bench inputs



Plant test: jumper PLANT_OUT -> EXT_SP (LOOP_SP 2–3) and AD3 W1 -> EXT_FB (LOOP_FB 2–3).

Power + decoupling



Analog PID quad flight controller

Sheet: /PID_Roll/
File: pid_roll.kicad_sch

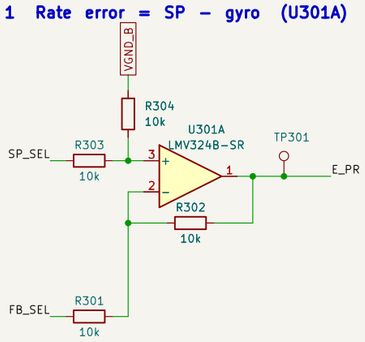
Title: Roll axis PID

Size: A3 | Date: 2026-10-20 | Rev:
KiCad E.D.A. 10.0.3 | Id: 3/6

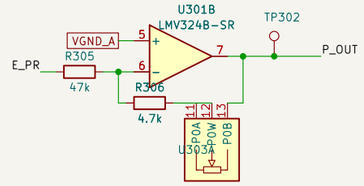
PITCH AXIS: rate PID + angle loop + bench plant (guide chips U4 = U301, U5 = U302)

Signals are deviations from VG = 2.048 V. Every P, I and D path is inverted twice. Gains are set by MCP4451 rheostat channels (A tied to W; R_BW = code/256 x 100k).

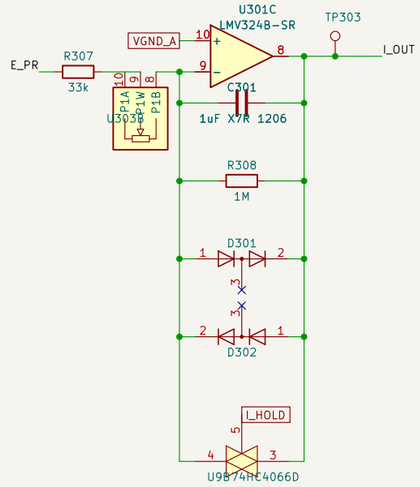
1 Rate error = SP − gyro (U301A)



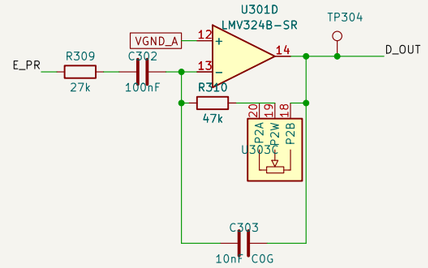
2 P, inverting (U301B)



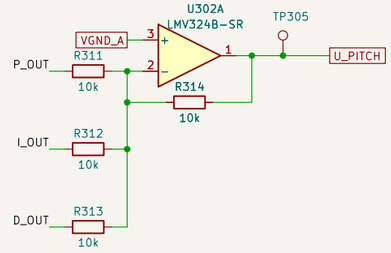
3 Leaky I + clamp + reset (U9 = 74HC4066) (U301C)



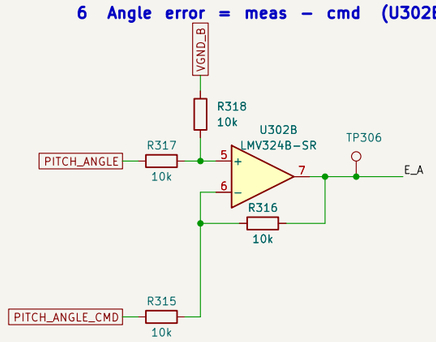
4 Filtered D (U301D)



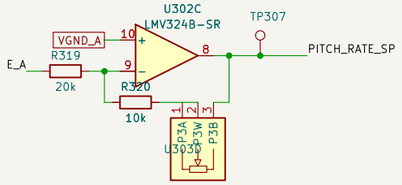
5 Summer -> U_PITCH (U302A)



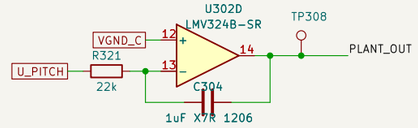
6 Angle error = meas − cmd (U302B)



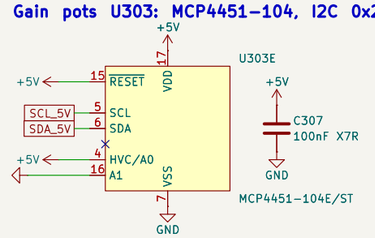
7 Angle P -> rate SP (U302C)



8 Bench plant (U302D)

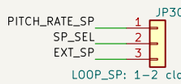


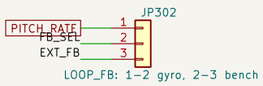
Gain pots U303: MCP4451-104, I2C 0x2D (A1=0, A0=1)



ch0 = P: code 128 = 50k (Kp 1.16)
ch1 = I: code 171 = 67k (Ki 10/s)
ch2 = D: code 128 = 50k (Kd 9.7 ms)
ch3 = Angle P: code 64 = 25k (Go 1.75)
Power-on = code 128 (mid); the shim ATtiny writes the stored codes over I2C at boot.

Loop select + bench inputs

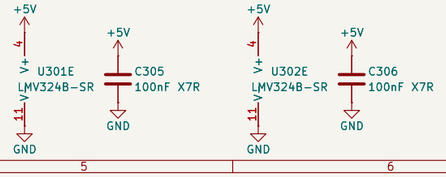






Plant test: jumper PLANT_OUT -> EXT_SP (LOOP_SP 2–3) and AD3 W1 -> EXT_FB (LOOP_FB 2–3).

Power + decoupling



Analog PID quad flight controller

Sheet: /PID_Pitch/
File: pid_pitch.kicad_sch

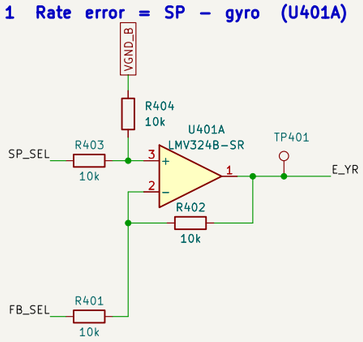
Title: Pitch axis PID

Size: A3 | Date: 2026-10-20 | Rev:
KiCad E.D.A. 10.0.3 | Id: 4/6

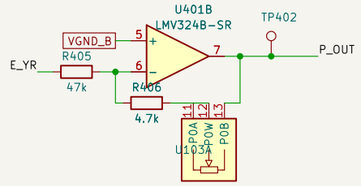
YAW AXIS: rate PID (guide chips U6 = U401, U7 = U402)

Signals are deviations from VG = 2.048 V. Every P, I and D path is inverted twice. Gains are set by MCP4451 rheostat channels (A tied to W; R_BW = code/256 x 100k).

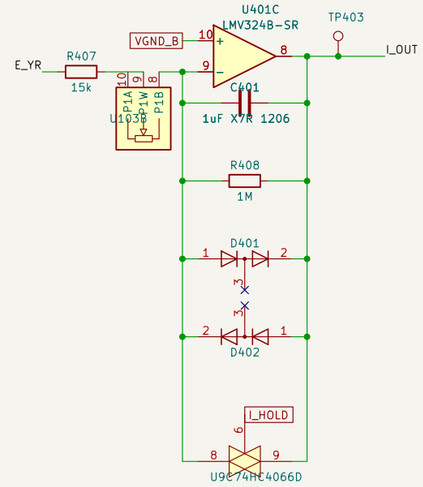
1 Rate error = SP − gyro (U401A)



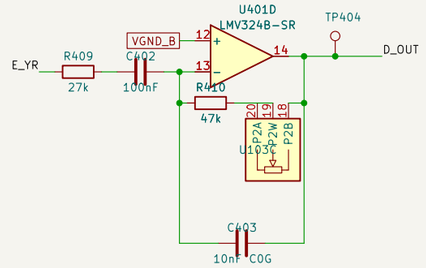
2 P, inverting (U401B)



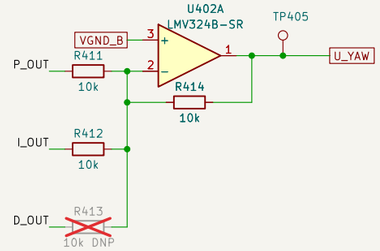
3 Leaky I + clamp + reset (U9 = 74HC4066) (U401C)



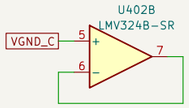
4 Filtered D (U401D)



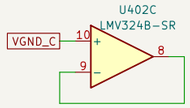
5 Summer -> U_YAW (U402A)



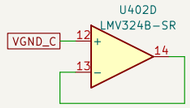
Spare follower (U402B)



Spare follower (U402C)



Spare follower (U402D)



Gain pots U103: MCP4451-104, I2C 0x2E (A1=1, A0=0)

ch0 = P: code 128 = 50k (Kp 1.16)

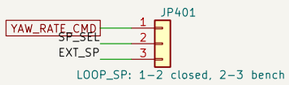
ch1 = I: code 128 = 50k

ch2 = D: code 128 = 50k (Kd 9.7 ms)

ch3 = VSET trim (reference sheet)

Power-on = code 128 (mid); the shim ATtiny writes the stored codes over I2C at boot.

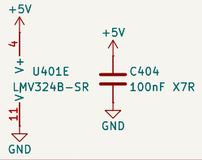
Loop select + bench inputs

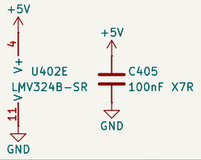






Power + decoupling





Analog PID quad flight controller

Sheet: /PID_Yaw/
File: pid_yaw.kicad_sch

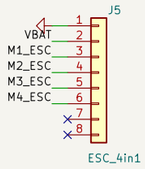
Title: Yaw axis PID

Size: A3 | Date: 2026-10-20 | Rev:
KiCad E.D.A. 10.0.3 | Id: 5/6

CONTROL I/O: ESC harness + power, mixer/ESC-driver ATtiny, integrator-reset switch

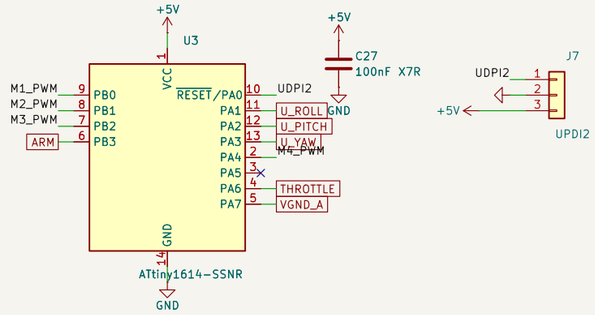
Shim ATtiny (root) parses CRSF, drives ARM and I_HOLD. This ATtiny reads the PID outputs, mixes quad-X and drives the 4 ESC inputs.

ESC: Aero Selfie 4in1 30A, 8-pin harness (cut one end, solder here)

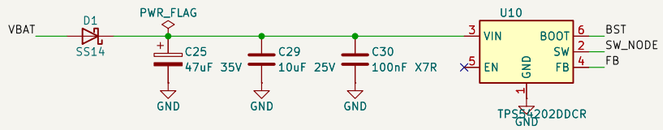


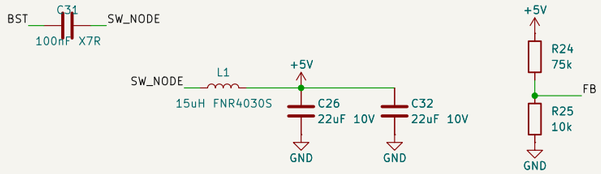
J5 order per ESC leaflet: 1 BAT-, 2 BAT+, 3-6 motor 1-4 signal, 7 no output, 8 current ADC. Check pin 1 = ESC minus pad with a DMM.

Mixer + ESC driver: ATtiny1614 #2 (ADC in, 4x PWM / OneShot / DShot out)



Power: VBAT (2S-4S via ESC harness) -> SS14 -> TPS54202 buck (500 kHz) -> +5V



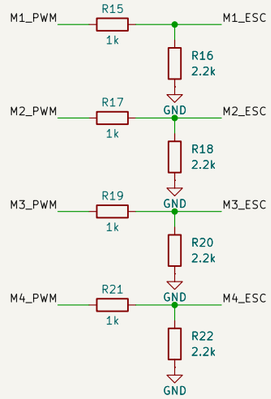


TPS54202: 4.5-28 V in, 2 A, Eco-mode at light load. Keep SW_NODE copper small and away from the PID sheets; C29/C30 and the GND pin within 2 mm of U10.

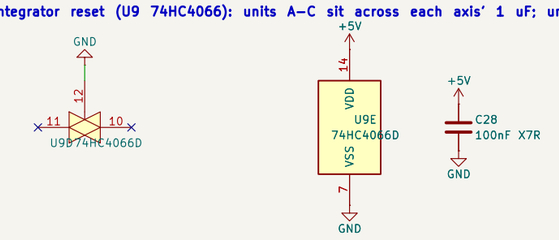


Firmware notes: ESC idle/arm pulse 800-1050 us, run 1050-2000 us (ESC leaflet). Disarmed: hold all outputs at 1000 us and I_HOLD high.

ESC signal: 1k / 2.2k divider -> 3.4 V high (ESC MCU may not be 5 V tolerant)



Integrator reset (U9 74HC4066): units A-C sit across each axis' 1 uF; unit D spare



Analog PID quad flight controller		
Sheet: /Control_IO/ File: control_io.kicad_sch		
Title: Control I/O: ESC, power, mixer		
Size: A3	Date: 2026-10-20	**Rev:**
KiCad E.D.A. 10.0.3		Id: 6/6